

Week 12 • Time-tracking, and the idea of context

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Latest version: <https://itu-tid.github.io/lecture-notes-pdf/Week-12.pdf>

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Field-Level Permissions, and the Two-Table Workaround

Source: [Lectures/Technical/Backend/Field-Level-Permissions.md](#) — something unclear? [Open an issue](#), or send a pull request.

In [Authentication and Authorization \(in Parse\)](#) every permission we set applied to a whole object. That is the only granularity Parse offers, and sooner or later you want less than that.

ACL does not work at the field level

In Parse, **ACLs work at the object level, not at the field/column level**. You cannot make some fields public and other fields private within the same object.

The example that forces the issue:

- You want everyone to **see your to-do's task name** and done status (public read).
- But you want to **keep your time-tracking data private** (only you can read it).

One object, two audiences. There is no ACL that expresses it.

Problem: you cannot set some fields private and some public

```
// This does NOT work - you can't set different permissions per field
const todo = new TodoItem();
todo.set("text", "Write report");           // want this PUBLIC
todo.set("done", false);                   // want this PUBLIC
todo.set("totalTime", 3600000);             // want this PRIVATE
todo.set("currentSessionStart", new Date()); // want this PRIVATE

const acl = new Parse.ACL(currentUser);
acl.setPublicReadAccess(true); // This makes ALL fields public!
todo.setACL(acl);
```

The ACL is a property of the row, not of any column in it. `setPublicReadAccess(true)` opens every field at once, `totalTime` along with `text`.

Solution: split the data into two tables with a 1-to-1 relationship

If the permissions differ, the objects differ. Put the public fields in one class and the private ones in another, and point the second at the first.

Table 1: TodoItem (public)

```

const TodoItem = Parse.Object.extend("TodoItem");
const todo = new TodoItem();
todo.set("text", "Write report");
todo.set("done", false);
todo.set("owner", currentUser);

// Public read, owner write
const acl = new Parse.ACL(currentUser);
acl.setPublicReadAccess(true);
todo.setACL(acl);
await todo.save();

```

Table 2: TodoTimeTracking (private)

```

const TodoTimeTracking = Parse.Object.extend("TodoTimeTracking");

const timeTracking = new TodoTimeTracking();

timeTracking.set("todoId", todo); // Pointer to the TodoItem
timeTracking.set("totalTime", 3600000);
timeTracking.set("currentSessionStart", new Date());
timeTracking.set("owner", currentUser);

// Private - only owner can read and write
const acl = new Parse.ACL(currentUser);
timeTracking.setACL(acl);
await timeTracking.save();

```

The pointer is the same mechanism as any other one-to-one relationship — see [Relationships Between Object Classes](#). What is new is the *reason* for the split: not that the data is a separate thing, but that it needs a separate ACL.

Exam Questions

1. Why can't ACLs be set at the field level, and what's the workaround?
2. You split a to-do's private time-tracking into its own class. What stops another user reading it, given they can already read the `TodoItem` it points at?

Something unclear in this section? Open an issue.